4 Fig. 4.1 shows an electron with a horizontal velocity of 5.0×10^{7} m s^{-1} entering a region midway between two flat parallel metal plates, each of length 12.0 cm. The plates are separated by a distance of 2.8 cm.

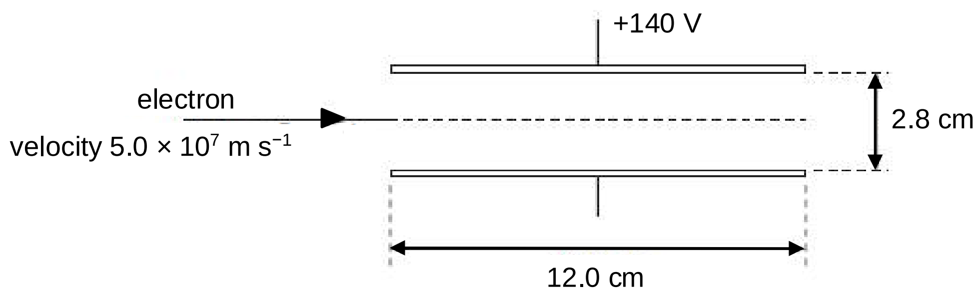

Fig. 4.1

The space between the plates is a vacuum. An upward electric field of field strength 1.40×10^{4} N C^{-1} may be assumed to be uniform in the region between the plates and zero outside this region. The potential of the top plate is +140 V.

(a) Determine potential of the bottom plate.

potential = …………………………….... V [2]

(b) For the electron between the plates,

(i) show that the magnitude of the acceleration is 2.5×10^{15} m s^{-2},

[1]

(ii) determine the time to travel a horizontal distance equal to the length of the plates.

time = ……….………….……... s [2]

(c) Use your answers in **(b)** to determine whether the electron will hit one of the plates or emerge from between the plates.

………………………………………………………………………………………………..

……………………………………………………………………………………………….. [3]

5 An electron enters a region R perpendicularly to a uniform magnetic field which is directed